

KG-BiLM: Knowledge Graph Embedding via Bidirectional Language Models

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Abstract

Recent advances in knowledge representation learning (KRL) highlight the urgent necessity to unify symbolic knowledge graphs (KGs) with language models (LMs) for richer semantic understanding. However, existing approaches typically prioritize either graph structure or textual semantics, which leaves a gap, i.e., a unified framework that simultaneously captures global KG connectivity, nuanced linguistic context, and discriminative reasoning semantics. To bridge this gap, we introduce **KG-BiLM**, a bidirectional LM framework that fuses structural cues from KGs with the semantic expressiveness of generative transformers. KG-BiLM incorporates three key components: (i) *Bidirectional Knowledge Attention*, which eliminates the causal mask to enable full interaction among all tokens and entities; (ii) *Knowledge-Masked Prediction*, which encourages the model to leverage both local semantic contexts and global graph connectivity; and (iii) *Contrastive Graph Semantic Aggregation*, which preserves KG structure via contrastive alignment of sampled subgraph representations. Extensive experiments on standard benchmarks demonstrate that KG-BiLM¹ outperforms the state-of-the-art baselines in link prediction, especially on large-scale graphs with complex multi-hop relations—validating its effectiveness in unifying structural information and textual semantics.

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¹The source code of KG-BiLM is available at <https://github.com/zirui-chen/kg-bilm>



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CCS Concepts

• **Computing methodologies** → **Knowledge representation and reasoning.**

Keywords

Knowledge Graph, Language Model, Representation Learning, Embedding

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1 Introduction

Recent advances in knowledge representation learning (KRL) highlight the need for unified approaches that integrate large-scale knowledge graphs (KGs) with structural representations and modern language models (LMs) with semantic understanding [31] [28]. In particular, significant progress in both symbolic and neural paradigms [15] has motivated research beyond traditional knowledge graph embedding (KGE) methods to leverage contextual semantics more fully [5] [17]. Moreover, the convergence of symbolic reasoning and neural text understanding has become increasingly prominent [53]. Consequently, investigating hybrid approaches that preserve KG structural relationships while capturing deep linguistic information is critical [38] [32].

Despite extensive research on KRL, the field remains fragmented. As shown in Fig.1, KGE methods, such as translation-based and semantic matching models, operate solely on symbolic triples and

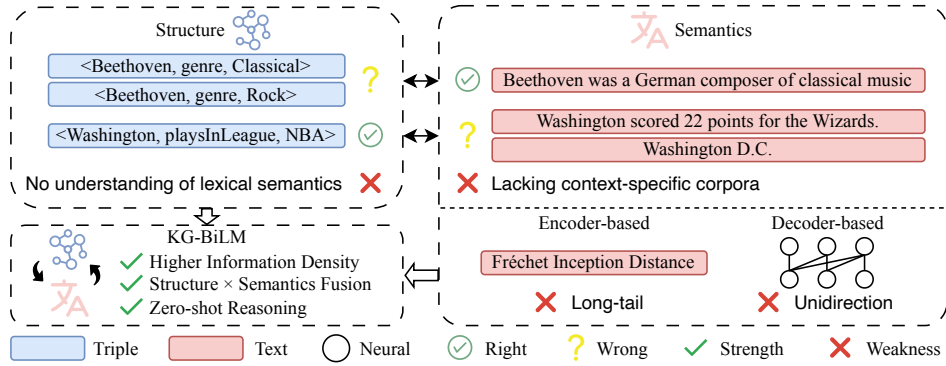


Figure 1: Motivating landscape and complementary failure modes in KRL. (a) Translation-based models excel at structural generalization but often underuse textual descriptions; subsequent variants (e.g., TransH/TransR) alleviate 1–N issues yet remain text-agnostic. (b) Pure encoders struggle on long-tail entities with scarce text. (c) Decoder-only LMs are uni-directional, limiting tight structure–semantics fusion. (d) KG-BiLM addresses these gaps via a unified *bidirectional* architecture that fuses KG links and associated texts at the token level.

thus fail to capture lexical semantics, severely limiting their adaptability to heterogeneous downstream tasks [34]. Encoder-based transformers (e.g., BERT [11]) inject contextual word representations but are typically pre-trained on generic corpora, lacking the domain-specific text required for KG reasoning. Consequently, they struggle with long-tail entities and relations and cannot leverage the richer generative cues available to decoder-only models [3]. Conversely, applying decoder-based large language models (e.g., LLaMA [37]) directly to KG tasks introduces new challenges. The inherent unidirectional generation paradigm obscures bidirectional dependencies and often ignores explicit structural constraints essential for faithful inference [56].

To bridge this research gap, we present a hybrid architecture called *KG-BiLM*, aiming to (i) achieve higher information density than conventional KG encoders, (ii) integrate linguistic cues into KG representations, and (iii) enable zero-shot encoding capabilities. By overcoming the longstanding trade-offs between bidirectional structural reasoning, local semantic fidelity, and global topological consistency, *KG-BiLM* unifies the strengths of both structure- and semantics-oriented representation learning.

KG-BiLM unifies structural and semantic information via three core techniques: (1) **Bidirectional Knowledge Attention** removes the causal mask from the decoder so that each token can attend to both past and future contexts, thereby strengthening inter-triple connections; (2) **Knowledge-Masked Prediction** selectively masks triple and lexical tokens, forcing the model to leverage local lexical context alongside the global graph structure to reconstruct missing tokens, thereby integrating rich linguistic cues into the KG embeddings; and (3) **Contrastive Graph Semantic Aggregation** applies contrastive learning on multiple graph-sampled views, aligning structural and semantic embeddings to preserve topology, enhance discriminative power, and facilitate zero-shot encoding of novel entities and relations. Together, these innovations transform a decoder-only transformer into a unified framework that jointly harnesses global KG structure and rich linguistic context for more accurate and generalizable reasoning.

The main contributions of this paper are as follows:

- **Global Connectivity:** A novel causal masking scheme that enables bidirectional attention between tokens and entities along graph edges. This enhances multi-hop relational reasoning and captures long-range dependencies in a joint structural–textual context, thereby improving knowledge propagation with minimal overhead while preserving generative capabilities.
- **Contextual Inference:** An adaptive masking-recovery mechanism that hides and reconstructs entities and tokens using both graph adjacency and linguistic cues. This enforces structural–linguistic synergy, supports zero-shot generalization and robust missing-information recovery, and leverages a position-shifted loss to blend global and local signals effectively.
- **Semantic Discrimination:** A contrastive learning objective to align semantically similar graph–text pairs and repel dissimilar ones. This sharpens global embedding boundaries, ensures cluster cohesion and distinctive entity semantics at scale, and enhances retrieval precision under noisy, heterogeneous subgraph augmentations.
- **Comprehensive Evaluation:** We empirically demonstrate that *KG-BiLM* effectively preserves graph-structural cues while integrating textual semantics, establishing a new paradigm for holistic knowledge representation learning.

2 Related Work

2.1 Knowledge Graph Embeddings

Recent KGE research has moved beyond early translational families to more expressive geometries and inductive designs that strive for stronger reasoning and generalisation. Nonetheless, purely structural models still struggle to capture contextual semantics and to operate in open-world settings. ExpressiveE [33] embeds entities as points and relations as parallelograms in a learnable Euclidean subspace, enabling the model to reproduce relation patterns such as hierarchy or composition with elegant geometric operations. TGraIL [23] combines sub-graph structure and type information to support

inductive link prediction where novel entities appear only at test time, while TaKE [14] injects latent type vectors into any backbone KGE model and devises a type-aware negative-sampling strategy to stabilise training. Although the above methods markedly improve structural fidelity, they inherit two long-standing weaknesses: (i) scarce usage of contextual semantics (e.g., textual descriptions) means that zero-shot prediction remains brittle; (ii) most models must be retrained whenever a new graph is introduced, hampering scalability. KG-BiLM retains the high information density of geometric encoders yet enriches the embeddings with bidirectional linguistic cues supplied by a large language model (LLM). In doing so, the framework alleviates the semantic sparsity and domain-adaptation barriers observed above.

These studies highlight the need for more flexible architectures that can effectively handle heterogeneous downstream scenarios that increasingly extends beyond entity-level embedding precision to more generalizable and context-sensitive representations.

2.2 Encoder-Based KRL

Recent encoder-based KRL treats KG triples as natural language to inject semantics that purely geometric models miss. Three compact strands dominate. *Triple-based* encoders linearise (h, r, t) and score the $[\text{CLS}]$ vector: KG-BERT [51] established the recipe; MTL-KGC [19] adds auxiliary tasks to expose KG-specific bias; K-BERT [24] softly grafts triples into the input tree; MLMML [8] reframes link prediction as MLM for open-vocabulary completion; promptable variants like PKGC and CSProm-KG [6, 27] reduce parameter updates via cloze templates and learned soft prompts. *Translation-based* encoders mimic TransE while keeping PLM expressiveness: StAR [42] aligns (h, r) and t in a shared metric space with margin/contrastive losses, while SimKGC and LP-BERT [21, 43] scale contrastive learning with rich negative pools and masked entity–relation objectives for inductive generalisation. *Independent* encoders disentangle components: KEPLER [45] couples translational distance with Wikipedia MLM to steady long-tail entities; BERT-ResNet [26] adds residual CNNs to capture higher-order signals; BLP [9] builds local mini-graphs for unseen-entity reasoning.

Despite steady gains, two weaknesses persist: (i) heavy reliance on local windows and handcrafted negatives can miss broader semantics and make zero-shot brittle; (ii) most models still require extensive hard-negative mining when the graph domain shifts. KG-BiLM retains the compactness of geometric encoders while enriching them with bidirectional linguistic cues from a large language model, alleviating semantic sparsity and easing cross-graph adaptation.

2.3 Decoder-Based KRL

Decoder-based KRL leverages autoregressive LLMs (e.g., LLaMA-2 [37], GPT-4 [30]) to generate or evaluate textual surrogates of triples with minimal tuning. Three concise paradigms have emerged. *Description-generation* synthesises missing context for long-tail entities: Contextualisation Distillation (CD) [22] prompts an LLM for paragraph-level evidence and distils it into a student via reconstruction/generation losses; CP-KGC [50] constrains templates with ontology slots to curb drift. *Prompt-engineering* casts KGC as QA: KG-LLM [52] scores plausibility via yes/no prompts, while

KICGPT [46] adds structure-aware in-context examples at inference to improve long-tail MRR without gradient updates. *Structural fine-tuning* injects explicit KG signals into decoders: KoPA [57] converts pre-trained TransE-style vectors into prefix tokens through a light-weight adapter, and KG-GPT2 [2] uses perplexity over linearised triples; hybrid pipelines such as RESKGC [54] combine retrieval, generation, and structural re-ranking.

Yet, these methods typically output text without preserving a durable, graph-aligned representation, and they inherit decoder costs. By coupling high-density geometric encodings with bidirectional language guidance, KG-BiLM keeps structural integrity front-and-centre while importing the contextual breadth of LLMs, improving robustness in zero-/few-shot and cross-domain settings.

3 The KG-BiLM Model

This section introduces KG-BiLM, a knowledge graph-enhanced bidirectional large language model that unifies structural and textual semantics for robust knowledge representation.

3.1 Preliminaries

Let $\mathcal{G} = (\mathcal{E}, \mathcal{R}, \mathcal{T})$ be a knowledge graph, where $\mathcal{E}$ is the set of entities, $\mathcal{R}$ the set of relations, and $\mathcal{T} \subseteq \mathcal{E} \times \mathcal{R} \times \mathcal{E}$ the set of triplets (e_i, r, e_j) describing relation $r \in \mathcal{R}$ between $e_i, e_j \in \mathcal{E}$. From pre-training corpora we derive a textual vocabulary $\mathcal{V}$. Our goal is to learn embeddings in a shared d -dimensional space so that every entity $e \in \mathcal{E}$ and every token $v \in \mathcal{V}$ is represented in $\mathbb{R}^d$, preserving both graph structure and linguistic context. Given a tokenized input sequence $\mathbf{x} = (x_1, \dots, x_N)$, which may intermix entity and relation names and natural-language tokens, the model parameters Θ map $\mathbf{x}$ to an initial embedding matrix $\mathbf{H}^{(0)}$. The full list of symbols is provided in Appendix A.

3.2 Overview Framework

Figure 2 presents the overall architecture. KG-BiLM first encodes KG triples and entity description into a unified token sequence. These tokens are then processed by a Bidirectional Knowledge Attention module, which injects graph-aware masks. Next, we apply a Knowledge-Masked Prediction objective to reconstruct the masked tokens, leveraging in-batch negatives for more effective contrastive learning. Finally, a Contrastive Graph Semantic Aggregation step aligns paired views via an InfoNCE loss. The functions of the three modules are as follows:

- **Bidirectional Knowledge Attention (BKA)** revises the causal mask of a decoder-style transformer so that every position may attend to every other, strengthening entity–entity and token–token interactions without sacrificing autoregressive capabilities.
- **Knowledge-Masked Prediction (KMP)** strategically hides entities and textual tokens and trains the model to recover them using both the global graph topology and local linguistic context, thereby encouraging rich cross-modal reasoning.
- **Contrastive Graph Semantic Aggregation (CGSA)** samples semantically related sub-graphs or textual variants and pulls their representations closer while pushing apart dissimilar samples, sharpening decision boundaries in the embedding space.

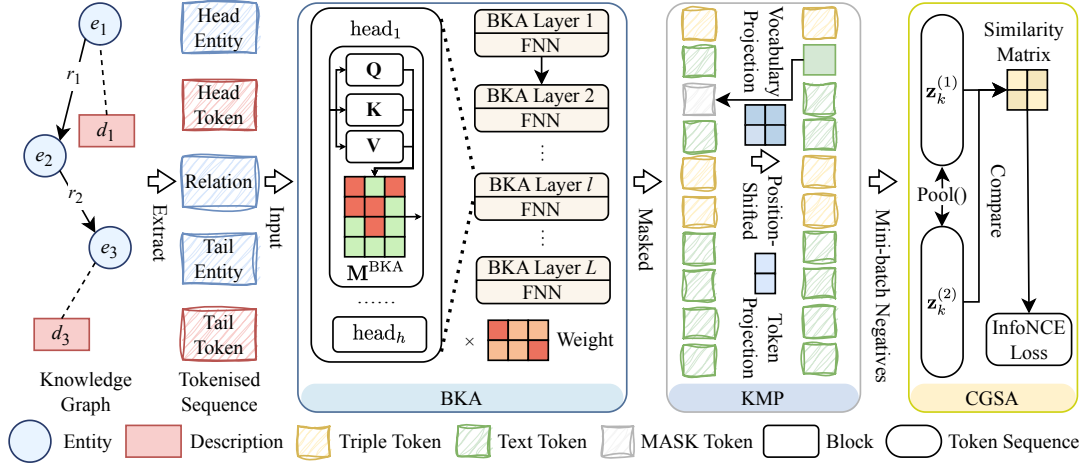


Figure 2: Overview of the KG-BiLM Architecture. A triple (and its neighborhood) is linearized as *head name* → *head description* → *relation* → *tail name* → *tail description*. BKA applies graph-aware attention masks to unify sequential and structural views. KMP masks entity-/relation-critical tokens and reconstructs them to couple local text and global KG context. CGSA samples structure-preserving subgraph views and aligns their sentence-level embeddings via contrastive loss, enhancing generalization and interpretability.

3.3 Bidirectional Knowledge Attention

Existing KGE methods typically rely on unidirectional or strictly local attention, which can hinder the learning of coherent global embeddings. Encoder-based approaches (e.g., BERT-like architectures) employ fully bidirectional attention but often lack the generative expressivity required for open-ended inference. Decoder-based transformers, by contrast, use a causal mask that blocks access to future tokens, making it difficult to capture symmetric relations or long-range entity co-occurrences. Consequently, a refined attention scheme is required to bridge these limitations.

Formulation of Bidirectional Knowledge Attention. Formally, Let L be the number of transformer layers, and let $\mathbf{H}^{(\ell)} \in \mathbb{R}^{N \times d}$ be the hidden states at the ℓ -th layer. For layer ℓ , define the query, key, and value matrices as:

$$\mathbf{Q} = \mathbf{H}^{(\ell)} \mathbf{W}_Q, \quad \mathbf{K} = \mathbf{H}^{(\ell)} \mathbf{W}_K, \quad \mathbf{V} = \mathbf{H}^{(\ell)} \mathbf{W}_V \quad (1)$$

where $\mathbf{W}_Q, \mathbf{W}_K, \mathbf{W}_V \in \mathbb{R}^{d \times d_h}$ are trainable and d_h denotes the dimension of each attention head (so that $h \times d_h = d$ for h heads). In standard self-attention the mask $\mathbf{M}$ is set to $-\infty$ for positions (i, j) if $j > i$ in a strictly causal decoder, or 0 otherwise for a fully bidirectional encoder. Under the proposed BKA, a specialized mask $\mathbf{M}^{\text{BKA}}$ is introduced:

$$\mathbf{M}_{i,j}^{\text{BKA}} = \begin{cases} 0, & \text{if tokens } (x_i, x_j) \text{ may interact} \\ -\infty, & \text{otherwise} \end{cases} \quad (2)$$

The criterion “may interact” is determined not only by token positions but also by graph-based adjacency. Concretely, if x_i and x_j are entity tokens within the same sub-graph or share a relation path within a certain hop threshold, $\mathbf{M}_{i,j}^{\text{BKA}} = 0$. This ensures that each token may attend both backward and forward in the sequence, as well as to any closely connected entity in the KG.

Attention weights are then computed as:

$$\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}; \mathbf{M}^{\text{BKA}}) = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d_h}} + \mathbf{M}^{\text{BKA}}\right) \mathbf{V} \quad (3)$$

By permitting unrestricted forward and backward dependencies—while still enforcing structural constraints from the KG—the model captures richer, multi-hop relational patterns. Consequently, BKA seamlessly fuses textual context with graph connectivity, alleviating the data sparsity often encountered in traditional KGE methods.

Layer Update and Enhanced Representations. Denote the output of the ℓ -th layer by $\mathbf{H}^{(\ell+1)}$. Employing multi-head attention, we obtain:

$$\tilde{\mathbf{H}}^{(\ell)} = \text{Concat}(\text{head}_1, \text{head}_2, \dots, \text{head}_h) \mathbf{W}_O \quad (4)$$

where

$$\text{head}_i = \text{Attention}(\mathbf{Q}_i, \mathbf{K}_i, \mathbf{V}_i; \mathbf{M}^{\text{BKA}}) \quad (5)$$

and $\mathbf{W}_O \in \mathbb{R}^{(h \cdot d_h) \times d}$. Then fuse the attended representations with a residual connection and layer normalization:

$$\tilde{\mathbf{H}}^{(\ell)} = \text{LayerNorm}\left(\mathbf{H}^{(\ell)} + \tilde{\mathbf{H}}^{(\ell)}\right) \quad (6)$$

Next, a position-wise feed-forward network (FFN) is employed:

$$\mathbf{H}^{(\ell+1)} = \text{LayerNorm}\left(\tilde{\mathbf{H}}^{(\ell)} + \text{FFN}(\tilde{\mathbf{H}}^{(\ell)})\right) \quad (7)$$

This iterative transformation processes the sequence holistically, thereby allowing for the accumulation of signals from all tokens and their interlinked entities. The second advantage of using the multivariate approach is its capability to encode complex relational structures in a single pass, enhancing representational power for multiple downstream tasks.

By iterating this sequence of graph-aware attention, residual fusion, and feed-forward updates, the model holistically aggregates

signals from all tokens and their linked entities. This multivariate update not only captures complex relational structures in one pass but also substantially enhances the embedding power for downstream tasks.

3.4 Knowledge-Masked Prediction

Existing KRL pipelines often focus on direct link prediction or pairwise entity classification, which limits their ability to capture nuanced textual cues. In contrast, our method utilises knowledge-masked prediction (KMP) paradigm to gain a more holistic understanding of entity semantics. While KMP resembles traditional masked language modeling, it is specifically designed to incorporate relational paths and entity dependencies from the knowledge graph. This design makes the model robust to partial or missing observations—a key advantage for downstream zero-shot generalization.

Notation for Masked Entities and Tokens. Let $\mathbf{x}^m = (x_1^m, x_2^m, \dots, x_N^m)$ be the input sequence where a selected subset $\mathcal{M} \subset \{1, \dots, N\}$ of positions is masked. Each masked position $i \in \mathcal{M}$ could correspond to either an entity token or a textual token, replaced by a special mask symbol:

$$x_i^m = \begin{cases} x_i, & \text{if } i \notin \mathcal{M} \\ \langle \text{mask} \rangle, & \text{if } i \in \mathcal{M} \end{cases} \quad (8)$$

We denote the masking ratio by $\gamma = \frac{|\mathcal{M}|}{N}$, which can be tuned to regulate the difficulty of the inference task.

Predicting Masked Tokens and Entities. Once the BKA is applied, the hidden states $\mathbf{H}^{(L)}$ at the final layer encapsulate both local lexical context and global KG structure. Let $\mathbf{h}_i \in \mathbb{R}^d$ denote the final hidden representation at position i . The model then projects $\mathbf{h}_i$ into a distribution over the vocabulary $\mathcal{V}$ extended with entity symbols:

$$p_{\Theta}(x_i | \mathbf{h}_i) = \text{softmax}(\mathbf{W}_P \mathbf{h}_i + \mathbf{b}_P) \quad (9)$$

where $\mathbf{W}_P \in \mathbb{R}^{|\mathcal{V}+\mathcal{E}| \times d}$ and $\mathbf{b}_P \in \mathbb{R}^{|\mathcal{V}+\mathcal{E}|}$ are projection parameters. For each $i \in \mathcal{M}$, the training objective focuses on predicting the true token or entity x_i based on the modified input $\mathbf{x}^m$.

Position-Shifted Loss Assignment. Beyond standard token reconstruction, KMP shifts supervision from position i to $i-1$ so that the model must integrate both left and right contexts while respecting graph connectivity. This design (i) reduces over-reliance on local n-grams, (ii) exposes relational edges to global attention, and (iii) better couples textual and structural cues. Our ablations (Sec. 4) show that removing KMP consistently degrades performance (e.g., lower MRR and Hits@k), corroborating the above motivations.

$$\mathcal{L}_{\text{KMP}} = - \sum_{i \in \mathcal{M}} \log p_{\Theta}(x_i | \mathbf{h}_{i-1}) \quad (10)$$

where $\mathbf{h}_{i-1}$ denotes the hidden representation from the immediately preceding position. This position-shifted KMP scheme compels the network to aggregate contextual information from both preceding and succeeding tokens, as well as linked entities. It offers three key benefits: it enforces consistency across consecutive

positions, encourages global attention to relational edges, and ensures that local neighborhoods in the sequence capture relevant semantically related tokens.

Link to Zero-Shot Capability. By repeatedly training the model to infer masked entity tokens with partial observations, it learns to exploit both cross-position and cross-relational dependencies. Consequently, the learned embeddings become adaptable to new entities and unseen textual patterns. The synergy between the BKA and the position-shifted KMP modules thus underpins robust zero-shot encoding: once structural and linguistic patterns are internalized, the model can generalize to novel contexts with minimal or no additional training data.

3.5 Contrastive Graph Semantic Aggregation

Despite the effectiveness of bidirectional attention and knowledge masking in unifying textual and graph-based cues, there remains a risk that entity embeddings become overly entangled with their local context and lose global distinctiveness. To mitigate this, we introduce a contrastive learning module—*Contrastive Graph Semantic Aggregation* (CGSA)—that preserves discriminative power while still benefiting from integrative representations. The contrastive objective naturally clusters semantically or relationally coherent samples and pushes apart incongruent pairs, thereby reinforcing the structural integrity of the knowledge-graph representation.

Sampling Mechanism and Dropout Variation. To achieve contrastive alignment, we draw two independent corrupted views $\mathbf{x}^{(1)}$ and $\mathbf{x}^{(2)}$ from the same original sequence or sub-graph. Each view is generated by applying a separate random dropout mask. Specifically, for a given sub-graph or textual snippet s , two corrupted views s_1 and s_2 are generated through random dropout or data augmentation, yielding token sequences $\mathbf{x}^{(1)}$ and $\mathbf{x}^{(2)}$. Both are passed through the aforementioned BKA and KMP modules, resulting in final embeddings:

$$\mathbf{z}^{(1)} = \text{Pool}(\mathbf{H}^{(L)}(\mathbf{x}^{(1)})), \quad \mathbf{z}^{(2)} = \text{Pool}(\mathbf{H}^{(L)}(\mathbf{x}^{(2)})) \quad (11)$$

where $\text{Pool}(\cdot)$ (e.g., mean pooling or a special classification token) aggregates the sequence-level embedding.

Contrastive Objective. Denote a minibatch by $\{\mathbf{z}_k^{(1)}, \mathbf{z}_k^{(2)}\}_{k=1}^B$, where each index k corresponds to a distinct pair derived from the same original sub-graph s_k . Let $\text{sim}(\mathbf{u}, \mathbf{v}) = \frac{\mathbf{u}^T \mathbf{v}}{\|\mathbf{u}\| \|\mathbf{v}\|}$ measure the cosine similarity. Inspired by InfoNCE [40], we formulate the CGSA loss as:

$$\mathcal{L}_{\text{CGSA}} = - \sum_{k=1}^B \log \left(\frac{\exp(\text{sim}(\mathbf{z}_k^{(1)}, \mathbf{z}_k^{(2)}) / \tau)}{\sum_{\ell=1}^B \exp(\text{sim}(\mathbf{z}_k^{(1)}, \mathbf{z}_{\ell}^{(2)}) / \tau)} \right) \quad (12)$$

where τ is a temperature hyperparameter. The numerator encourages high similarity between the two views of the same entity-text pair, while the denominator discourages alignment with representations of other pairs in the minibatch. By maximizing the similarity of paired views and minimizing the similarity with respect to non-matching samples, CGSA promotes cluster cohesion for semantically similar embeddings and cluster separation for dissimilar embeddings.

Integration with Global KG Semantics. The CGSA excels at preserving global semantics in large-scale knowledge graphs with

complex relational patterns. By enforcing that embeddings remain discriminative under repeated dropout corruptions, it also fosters stable representations across paraphrases or reordered textual segments describing the same entity. For example, if two sub-graphs s_i and s_j share the majority of their entities but differ in minor textual descriptions, the contrastive loss anchors their embeddings to be closer than those of unrelated sub-graphs. In doing so, CGSA addresses a key limitation of earlier generative LM-based methods, wherein structural coherence was often overshadowed by purely textual factors.

Where does zero-shot ability come from? In KG-BiLM, zero-shot generalization emerges from the *complementarity* between KMP (inferring masked entity/relation tokens from local text plus structural context) and CGSA (stabilizing and aligning representations across semantically equivalent subgraph views).

4 Experiments

This section empirically evaluates KG-BiLM on four widely used knowledge graph benchmarks, comparing it against state-of-the-art symbolic, neural and hybrid alternatives. We first detail the experimental setup, then analyse performance on structure-only and

semantically-enriched datasets, followed by a component ablation, a zero-shot study, and a qualitative analysis.

4.1 Experimental Settings

Datasets. Table 1 summarises the statistics of the four benchmark datasets. Two of them (FB15k-237 and WN18RR) consist solely of structured triples without any accompanying textual entity descriptions, thus focusing on a model’s ability to learn graph structures. The other two (FB15k-237N and Wikidata5M) enrich each entity with detailed textual descriptions to facilitate semantic integration. Notably, FB15k-237N exhibits a pronounced long-tail distribution of entities, and the large-scale Wikidata5M further allows for zero-shot inference evaluation. Together, these richer datasets provide a more challenging testbed for evaluating our model’s capacity to fuse textual semantics, handle long-tail entities, and perform zero-shot reasoning.

Baselines. We evaluate two major paradigms: (1) KGE methods (TransE, DistMult, ComplEx, etc.) embedding entities and relations via distance- or multiplicative-scoring; (2) Transformer-based models (KG-BERT, SimKGC, KG-S2S, etc.) enriching KG embeddings with textual context, but may underutilize explicit graph structure.

Table 2: Summary of essential baseline link prediction metrics on WN18RR and FB15k-237.

Model	WN18RR					FB15k-237				
	MR	MRR	Hits@1	Hits@3	Hits@10	MR	MRR	Hits@1	Hits@3	Hits@10
TransE [4]	2300	24.3	4.3	44.1	53.2	223	27.9	19.8	37.6	47.4
DistMult [49]	3704	44.4	41.2	47.0	50.4	411	28.1	19.9	30.1	44.6
ComplEx [39]	3921	44.9	40.9	46.9	53.0	508	27.8	19.4	29.7	45.0
ConvE [10]	4464	45.6	41.9	47.0	53.1	245	31.2	22.5	34.1	49.7
TuckER [1]	–	47.0	44.3	48.2	52.6	–	35.8	26.6	39.4	54.4
CompGCN [41]	–	47.9	44.3	49.4	54.6	–	35.5	26.4	39.0	53.5
QuatDE [12]	1977	48.9	43.8	50.9	58.6	90	36.5	26.8	40.0	56.3
NBFNet [59]	–	55.1	49.7	–	66.6	–	41.5	32.1	–	59.9
GPT-4o [29]	99	1.1	0.0	0.1	0.2	99	1.4	0.3	0.5	0.6
KG-BERT [51]	97	21.6	4.1	30.2	52.4	153	23.7	16.9	26.0	42.7
Pretrain-KGE [58]	–	48.8	43.7	50.9	58.6	–	35.0	25.0	38.4	55.4
MPIKGC [48]	–	54.9	49.7	56.8	65.2	–	35.9	26.7	39.5	54.3
LaSS [35]	35	–	–	–	78.6	<u>108</u>	–	–	–	53.3
MKGL [13]	–	55.2	50	57.7	65.6	–	41.5	32.5	45.4	<u>59.1</u>
KG-S2S [7]	–	57.4	53.1	59.5	66.1	–	33.6	25.7	37.3	49.8
kNN-KGE [44]	–	57.9	52.5	–	–	–	28.0	37.3	–	–
CSPromp-KG [6]	–	57.5	52.2	59.6	67.8	–	35.8	26.9	39.3	53.8
CP-KGC [50]	–	<u>67.3</u>	<u>59.9</u>	<u>72.1</u>	80.4	–	33.8	25.1	36.5	51.6
SATKGC [20]	–	<u>67.3</u>	59.5	72.8	81.3	–	34.9	25.6	36.7	52.0
KG-FIT [18]	–	–	–	–	–	–	36.2	27.5	–	57.2
KICGPT [46]	–	56.4	47.8	61.2	67.7	–	41.2	<u>32.7</u>	44.8	55.4
KLR-KGC [16]	–	–	–	–	–	–	<u>41.4</u>	32.3	<u>44.9</u>	57.4
SimKGC [43]	–	66.7	58.8	<u>72.1</u>	<u>80.5</u>	–	33.6	24.9	36.2	51.1
KG-BiLM(Ours)	<u>67</u>	68.2	61.4	72.7	<u>80.5</u>	151	36.7	30.5	36.9	53.1

Table 1: Statistics of the KG benchmarks.

Dataset	#Ent.	#Rel.	Train	Valid	Test	Text
FB15k-237	14,541	237	272,115	17,535	20,466	No
WN18RR	40,943	11	86,835	3,034	3,134	No
FB15K-237N	13,104	93	87,282	7,041	8,226	Yes
Wikidata5M	4,594,485	822	20,614,279	5,163	5,133	Yes

Evaluation Metrics and Implementations. We evaluate our model using three widely adopted KRL metrics in a single framework: Mean Rank (MR) computes the average position of the true entity—lower values denote better overall ordering. Mean Reciprocal Rank (MRR) averages the inverse of each true-entity rank to

emphasize rapid retrieval, and Hits@ k ($k = 1, 3, 10$) reports the proportion of cases where the correct entity appears among the top- k predictions. KG-BiLM is *not* trained from scratch. We initialize from a pretrained decoder-only transformer and then adopt a two-stage paradigm: (1) knowledge-masked prediction (KMP) pretraining; (2) contrastive graph semantic aggregation (CGSA) finetuning; finally followed by task-specific finetuning on each KG dataset. All layers remain trainable so that bidirectional knowledge attention fully adapts to knowledge-centric reasoning.

4.2 Structural-Only Benchmark Results

Table 2 compares KG-BiLM with 18 competitive baselines on WN18RR and FB15k-237—two canonical datasets that omit textual descriptions to focus on pure graph reasoning. On WN18RR, KG-BiLM achieves the highest MRR, outperforming the previous best model, CP-KGC, by 0.009 MRR and matching SimKGC’s top Hits@10 score of 80.5. Notably, unlike CP-KGC and SimKGC, which rely on pre-trained text encoders, KG-BiLM attains superior accuracy using only symbolic triples. This improvement demonstrates the power of our BKA module: by removing the causal mask, the decoder conditions on full context both within and across triples, allowing multi-hop relationships to propagate more faithfully than in strictly sequential or bag-of-triples encoders.

On FB15k-237, the KGE model NBFNet still leads in absolute performance, but KG-BiLM remains competitive with recent Transformer-based baselines. We attribute the slightly lower MRR (relative to WN18RR) to FB15k-237’s larger relation vocabulary (237 vs. 11), which increases semantic ambiguity in the absence of textual cues. These results confirm that our graph-aware bidirectional decoding provides tangible benefits even without entity descriptions.

4.3 Semantically-Enriched Structural Benchmark Results

As Table 3 demonstrates, when we evaluate on datasets that augment each entity with natural-language descriptions, KG-BiLM sets new state-of-the-art results on both Wikidata5M and FB15k-237N. On the ultra-large Wikidata5M, our model achieves an MRR of 0.403, surpassing ReSKGC by 0.7 points and CSPromptKG by 2.3 points. Even more strikingly, KG-BiLM attains a Hits@1 score of 0.397—5.4 points higher than the previous best (ReSKGC). This gain stems directly from our KMP strategy: by randomly masking both entity and graph tokens during pre-training, the model learns to integrate semantically related but topologically distant evidence, which is crucial for handling textual spans that include unseen entity aliases or paraphrases.

Table 3: Summary of essential baseline link prediction metrics on Wikidata5M and FB15k-237N.

Model	Wikidata5M				FB15k-237N			
	MRR	Hits@1	Hits@3	Hits@10	MRR	Hits@1	Hits@3	Hits@10
TransE [4]	25.3	17.0	31.1	39.2	25.5	15.2	30.1	45.9
DistMult [49]	25.3	20.9	27.8	33.4	20.9	14.3	23.4	33.0
ComplEx [39]	30.8	25.5	—	39.8	24.9	18.0	27.6	38.0
RotatE [36]	29.0	23.4	32.2	39.0	27.9	17.7	32.0	48.1
QuatE [55]	27.6	22.7	30.1	35.9	—	—	—	—
ConvE [10]	—	—	—	—	27.3	19.2	30.5	42.9
CompGCN [41]	—	—	—	—	31.6	23.1	34.9	48.0
GPT-4o [29]	—	—	—	—	1.0	0.0	0.0	0.0
KG-BERT [51]	—	—	—	—	20.3	13.9	20.1	40.3
KG-S2S [7]	—	—	—	—	35.4	28.5	38.8	49.3
KEPLER [45]	21.0	17.3	22.4	27.7	—	—	—	—
SimKGC [43]	35.8	31.3	37.6	44.1	—	—	—	—
CSPrompt-KG [6]	38.0	34.3	39.9	<u>44.6</u>	36.0	28.1	39.5	51.1
ReSKGC [54]	<u>39.6</u>	<u>37.3</u>	<u>41.3</u>	43.7	—	—	—	—
CD [22]	—	—	—	—	<u>37.2</u>	<u>28.8</u>	<u>41.0</u>	<u>53.0</u>
KG-BiLM(Ours)	40.3	39.7	43.0	45.2	37.8	29.3	42.1	54.6

FB15k-237N exhibits the most pronounced long-tail entity distribution in our suite. Systems that rely on large-capacity text encoders often over-fit head entities and mis-rank rare ones. KG-BiLM mitigates this pitfall, outperforming the contrastive-pre-training approach CD on every metric, most clearly on Hits@10 (+1.6 points). The result confirms that CGSA scales gracefully: by aligning multiple graph-sampled views per mini-batch, KG-BiLM preserves

cluster cohesion for rare entities while simultaneously sharpening boundaries between semantically distinct neighborhoods. Ablation results (Section 4.5) back this claim quantitatively—the absence of CGSA drops Hits@10 on FB15k-237N by 1.5 points, the largest decrement among all variants.

Table 4: Ablation on validation split.

Variant	Wikidata5M		FB15k-237N	
	MRR	H@10	MRR	H@10
Full model	.403	.452	.378	.546
w/o BKA	.383	.426	.361	.525
w/o KMP	.390	.432	.366	.531
w/o CGSA	.397	.440	.370	.538

4.4 Ablation Study

Table 4 reports ablation results on the validation splits of Wikidata5M and FB15k-237N. Removing BKA incurs the largest performance drop on both datasets (−2.0 MRR on Wikidata5M; −1.7 MRR on FB15k-237N), confirming that global connectivity is the main contributor to enhanced multi-hop reasoning. Omitting KMP yields a somewhat smaller but still substantial degradation (−1.3 MRR on average), demonstrating that semantic–structural co-training remains indispensable even when textual information is abundant. Interestingly, excising CGSA hurts Hits@10 more than MRR, indicating that CGSA primarily boosts the model’s ability to rank challenging yet correct entities at the top, rather than merely improving coarse-grained ordering. Together, these controlled experiments validate that KG-BiLM’s gains stem from our architectural innovations rather than superficial scaling artifacts.

Table 5: Link prediction in zero-shot setting on Wikidata5M dataset.

Model	Wikidata5M			
	MRR	Hits@1	Hits@3	Hits@10
DKRL [47]	23.1	5.9	32.0	54.6
RoBERTa [25]	7.4	0.7	1.0	19.6
KEPLER [45]	40.2	22.2	51.4	73.0
SimKGC [43]	<u>71.4</u>	<u>50.9</u>	<u>78.5</u>	<u>91.7</u>
KG-BiLM (Ours)	74.8	53.7	81.6	93.8

4.5 Zero-shot Reasoning

Scope. Unless otherwise specified, our zero-shot evaluation follows a *within-KG* inductive protocol: test triples involve unseen entities from the *same* KG (e.g., Wikidata5M), and no cross-dataset checkpoint sharing is assumed. This differs from *cross-KG* setups that train a universal model and transfer to a *new* KG with unseen entities and relations.

The Zero-shot evaluation constitutes the most stringent test of generalization, since every evaluation triple contains at least one entity unseen during training, forcing models to extrapolate purely from textual descriptions. As shown in Table 5, KG-BiLM achieves an MRR of 0.748, outperforming SimKGC by 3.4 points and KEPLER by an impressive 34.6 points. These gains are consistent across the recall spectrum and culminate in a 2.1-point improvement on Hits@10. Qualitative analysis indicates that BKA remains pivotal: by leveraging future context to inform earlier predictions, the model can utilize downstream descriptions of unseen entities

and adjust preceding hypotheses—an ability lacking in causal decoders. Additionally, we observe that KMP enhances robustness to rare words by training the model to reconstruct missing entity names from partial mention cues, effectively serving as a denoising auto-encoder across both language and graph modalities.

Disentangling Structure and Language. Although KG-BiLM fuses both modalities in a shared bidirectional encoder, their training signals are decoupled (masked prediction versus contrastive alignment). We therefore evaluate two stress-tested variants: (i) *text-ablated* (remove all entity descriptions) and (ii) *structure-ablated* (detach KG masks in BKA). The former shows modest drops on sparse subgraphs, indicating that structural cues alone carry substantial predictive power; the latter hurts long-tail recovery more, underscoring the role of semantics in disambiguation.

4.6 Efficiency and Scalability

Hardware. All experiments run on 2× NVIDIA H100 (80GB) with NVLink.

Table 6: Dataset-level training cost.

Dataset	Total Steps	GPU Time (hrs)	Peak Memory (GB)
WN18RR	~31k	~3.2	~53
FB15k-237	~31k	~3.4	~55
FB15k-237N	~56k	~6.7	~63
Wikidata5M	~137k	~11.4	~78

Table 7: Component-wise overhead vs. gains (Wikidata5M).

Variant	ΔParams (M)	↑FLOPs	Peak Mem (GB)	Min/1k steps	GPU-hrs	ΔMRR
Baseline	–	–	88	3.6	34	0.373
+ BKA	+0.0	+11%	92	3.9	40	+0.020
+ KMP	+1.1	+6%	95	4.2	43	+0.013
+ CGSA	+3.6	+12%	104	4.9	52	+0.006
Full	+4.7	+28%	104	4.9	52	0.403

Overall, KMP offers the highest marginal utility per GPU-hour, while CGSA yields notable improvements in long-tail recall (see main-table Hits@10).

5 Conclusion

KG-BiLM advances knowledge representation by unifying symbolic graph structure with bidirectional linguistic modeling. By removing the causal mask, tokens can fully exploit global graph connectivity, while knowledge-masked prediction and contrastive graph–semantic aggregation bolster robustness to sparsity and enable zero-shot generalization. The resulting high-density embeddings combine structural fidelity with long-tail semantic coverage, outperforming standalone KGEs and LLMs across small-, large-scale, and long-tail benchmarks.

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A Notations

Table 8: Summary of notations used in this paper.

Notation	Explanation
$\mathcal{G} = (\mathcal{E}, \mathcal{R}, \mathcal{T})$	A knowledge graph consisting of entities $\mathcal{E}$, relations $\mathcal{R}$, and triples $\mathcal{T}$.
$\mathcal{E}$	Set of entities in the knowledge graph.
$\mathcal{R}$	Set of relations in the knowledge graph.
$\mathcal{T} \subseteq \mathcal{E} \times \mathcal{R} \times \mathcal{E}$	Set of relational triples (e_i, r, e_j) .
(e_i, r, e_j)	A triple linking head entity e_i to tail entity e_j by relation r .
$\mathcal{V}$	Textual vocabulary extracted from large-scale corpora.
d	Shared embedding dimension for all entities and tokens.
$e \in \mathcal{E}$	An individual entity.
$v \in \mathcal{V}$	An individual vocabulary token.
$\mathbf{x} = (x_1, \dots, x_N)$	Tokenised input sequence of length N .
N	Sequence length (number of tokens).
Θ	All trainable model parameters.
$\mathbf{H}^{(0)}$	Initial embedding matrix for the input sequence.
L	Number of stacked transformer layers.
h	Number of attention heads in multi-head attention.
d_h	Dimensionality of each attention head ($hd_h = d$).
$\mathbf{H}^{(\ell)} \in \mathbb{R}^{N \times d}$	Hidden states at transformer layer ℓ .
$\mathbf{Q}, \mathbf{K}, \mathbf{V}$	Query, key and value matrices in self-attention.
$\mathbf{W}_Q, \mathbf{W}_K, \mathbf{W}_V$	Projection parameters mapping hidden states to $\mathbf{Q}, \mathbf{K}, \mathbf{V}$.
$\mathbf{M}$	Standard attention mask (causal or fully bidirectional).
$\mathbf{M}^{\text{BKA}}$	Graph-aware bidirectional knowledge-attention mask (Eq. 2).
Attention($\cdot$)	Scaled-dot-product attention with mask $\mathbf{M}^{\text{BKA}}$.
$\tilde{\mathbf{H}}^{(\ell)}$	Concatenated multi-head attention output at layer ℓ .
$\mathbf{W}_O$	Output projection matrix after concatenating heads.
$\tilde{\mathbf{H}}^{(\ell)}$	Hidden states after residual connection and layer normalisation.
FFN($\cdot$)	Position-wise feed-forward network.
$\mathbf{H}^{(\ell+1)}$	Output hidden states of layer ℓ after FFN.
$\mathcal{M}$	Index set of masked positions in knowledge-masked prediction.
$\mathbf{x}^m$	Input sequence with tokens at $\mathcal{M}$ replaced by $\langle \text{mask} \rangle$.
$\gamma = \frac{ \mathcal{M} }{N}$	Masking ratio.
$\mathbf{h}_i$	Final hidden representation at position i .
$p_\Theta(x_i \mathbf{h}_i)$	Predicted distribution over tokens/entities for position i .
$\mathbf{W}_P, \mathbf{b}_P$	Output projection parameters for masked prediction.
$\mathcal{L}_{\text{KMP}}$	Knowledge-masked prediction loss (Eq. 10).
$\mathbf{x}^{(1)}, \mathbf{x}^{(2)}$	Two independently corrupted views of the same sample for contrastive learning.
$\mathbf{H}^{(L)}(\mathbf{x}^{(k)})$	Final-layer hidden states for view $k \in \{1, 2\}$.
$\mathbf{z}^{(1)}, \mathbf{z}^{(2)}$	Pooled sequence-level embeddings of the two views.
Pool($\cdot$)	Aggregation function (e.g. mean-pooling or [CLS] token).
B	Minibatch size for contrastive learning.
$\text{sim}(\mathbf{u}, \mathbf{v})$	Cosine similarity between vectors $\mathbf{u}$ and $\mathbf{v}$.

Algorithm 1: Bidirectional Knowledge Attention (BKA)

Input: Token sequence $\mathbf{x} = (x_1, \dots, x_N)$; knowledge graph $\mathcal{G} = (\mathcal{E}, \mathcal{R}, \mathcal{T})$; hop threshold h ; model parameters $\{\mathbf{W}_Q, \mathbf{W}_K, \mathbf{W}_V, \mathbf{W}_O\}$

Output: Contextually and structurally enriched representations $\mathbf{H}^{(L)}$

Initialize hidden states $\mathbf{H}^{(0)} \leftarrow \text{Embed}(\mathbf{x})$;

for $\ell \leftarrow 0$ **to** $L - 1$ **do**

for $i \leftarrow 1$ **to** N **do**

$\mathbf{q}_i \leftarrow \mathbf{H}_i^{(\ell)} \mathbf{W}_Q$;

for $j \leftarrow 1$ **to** N **do**

$\mathbf{k}_j \leftarrow \mathbf{H}_j^{(\ell)} \mathbf{W}_K$;

if $\text{CanInteract}(x_i, x_j, \mathcal{G}, h)$ **then**

$\mathbf{M}_{i,j}^{\text{BKA}} \leftarrow 0$;

else

$\mathbf{M}_{i,j}^{\text{BKA}} \leftarrow -\infty$;

$\alpha_i \leftarrow \text{softmax}(\frac{\mathbf{q}_i \mathbf{K}^\top}{\sqrt{d_h}} + \mathbf{M}_{i,:}^{\text{BKA}})$;

$\mathbf{V} \leftarrow \mathbf{H}^{(\ell)} \mathbf{W}_V$;

$\tilde{\mathbf{h}}_i \leftarrow \alpha_i \mathbf{V}$;

$\tilde{\mathbf{H}}^{(\ell)} \leftarrow \text{Concat}(\tilde{\mathbf{h}}_1, \dots, \tilde{\mathbf{h}}_N) \mathbf{W}_O$;

$\tilde{\mathbf{H}}^{(\ell)} \leftarrow \text{LayerNorm}(\mathbf{H}^{(\ell)} + \tilde{\mathbf{H}}^{(\ell)})$;

$\mathbf{H}^{(\ell+1)} \leftarrow \text{LayerNorm}(\tilde{\mathbf{H}}^{(\ell)} + \text{FFN}(\tilde{\mathbf{H}}^{(\ell)}))$;

return $\mathbf{H}^{(L)}$;

B Details of KG-BiLM Modules

B.1 Bidirectional Knowledge Attention

The BKA algorithm constitutes the connective tissue that fuses a Transformer’s bidirectional reasoning capacity with the multi-hop relational structure inherent in a KG. At its core, BKA generalises the classical attention mechanism by augmenting the binary attention mask with graph-aware semantics, thereby allowing each token to attend not only to every other token in its textual context but also to any entity token that is reachable by a bounded number of relational hops. The resulting attention topology is simultaneously sequence-complete and graph-selective, which contrasts starkly with the hard uni-directionality of decoder-only language models and the structure-agnostic bidirectionality of pure encoders.

B.1.1 Algorithmic Steps. Initial Embedding. The first line in the pseudocode instantiates $\mathbf{H}^{(0)}$ by concatenating standard token embeddings with position encodings and, crucially, KG-entity type embeddings. Unlike typical LM practice, entity IDs are not merely treated as sub-word strings; rather, they possess dedicated vectors that are updated jointly with lexical embeddings.

Graph-Aware Mask Construction. Within each layer ℓ , nested loops traverse token positions. For every pair (i, j) the routine computes $\mathbf{M}_{i,j}^{\text{BKA}}$. The helper CanInteract returns true if at least one of three conditions is satisfied: (1) $|i - j| \leq \delta$, where δ is a small local window to preserve micro-syntax; (2) x_i and x_j are verbal tokens and the model operates in a language-model mode; (3) their parent entities in the KG are connected by a path not exceeding h hops.

The final mask thus embodies both text contiguity and relational contiguity.

Attention Weight Computation. Once the mask matrix has been filled, the algorithm computes the scaled dot-product attention for each query. Because the mask contains $-\infty$ for disallowed pairs, the corresponding softmax weights vanish, enforcing hard structural sparsity. While the line-wise loops are explicit for clarity, real implementations vectorise all positions and heads, thereby preserving the $O(N^2)$ theoretical complexity of standard attention.

Residual Path and Feed-Forward Layers. After multi-head projection through $\mathbf{W}_O$, a residual connection and layer normalisation are applied. The FFN applies two linear maps with a gating non-linearity, typically GELU, and a residual link—a design that is empirically crucial for stabilising deep stacks.

B.1.2 Theoretical Properties. Symmetry Preservation. Because the mask allows i to see j and vice versa whenever CanInteract is true, relational symmetry (e.g., “married to”) can be internalised in a single pass rather than via post-hoc rule injection.

Multi-Hop Relational Reasoning. If the hop threshold $h > 1$, BKA implicitly permits the propagation of information along length- h paths. The self-attention coefficients can be shown to approximate a truncated power series of the KG adjacency matrix, thereby endowing the representations with spectral properties reminiscent of Graph Neural Networks.

Expressivity under Causal Decoding. Despite affording bidirectional flows, BKA does not sabotage the autoregressive nature required for text generation. During inference, tokens at position i only depend on already generated tokens plus KG neighbours, both of which are known at decoding time. Thus, BKA is compatible with beam search and other sampling schemes.

B.2 Knowledge-Masked Prediction

KMP extends the masked-language modelling paradigm by integrating explicit structural priors from the knowledge graph and, crucially, by shifting the prediction locus from the masked position to its immediate predecessor. This seemingly minor alteration endows KG-BiLM with a potent capacity for causal reasoning under partial observability, thereby promoting robust zero-shot transfer.

B.2.1 Algorithmic Steps. Mask Construction. A binary mask vector is sampled such that exactly γN positions are designated for replacement. For non-entity tokens, a uniform distribution suffices; for entity tokens, a weighted sampling scheme ensures that high-degree nodes are systematically challenged.

Encoding via BKA. The masked input $\mathbf{x}^m$ is fed through the BKA stack, which already embodies graph-informed attention patterns. Therefore, even though certain entity IDs have been replaced by an opaque mask token, their neighbours can still convey hints about their identity through multi-hop edges.

Cross-Entropy Accumulation. For each masked index i , logits are produced from $\mathbf{h}_i - 1$ and normalised via softmax over the joint vocabulary $\mathcal{V} \cup \mathcal{E}$. Crucially, the inclusion of entity symbols means the classifier is solving a mixed-type prediction problem. The average negative log-likelihood across all masked positions yields $\mathcal{L}_{\text{KMP}}$.

Algorithm 2: Knowledge-Masked Prediction (KMP)

Input: Token sequence $\mathbf{x} = (x_1, \dots, x_N)$; mask ratio γ ;
 projection parameters $\{\mathbf{W}_P, \mathbf{b}_P\}$; BKA-augmented
 Transformer

Output: Masked-prediction loss $\mathcal{L}_{\text{KMP}}$

Sample mask set $\mathcal{M} \subseteq \{1, \dots, N\}$ with $|\mathcal{M}| = \gamma N$;

foreach $i \in \mathcal{M}$ **do**
 Replace x_i with $\langle \text{mask} \rangle$ to obtain $\mathbf{x}^m$

$\mathbf{H}^{(L)} \leftarrow \text{BKA_Forward}(\mathbf{x}^m)$;
 $\mathcal{L}_{\text{KMP}} \leftarrow 0$;

foreach $i \in \mathcal{M}$ **do**
 $\mathbf{h}_{i-1} \leftarrow \mathbf{H}_{i-1}^{(L)}$;
 $\ell_i \leftarrow \mathbf{W}_P \mathbf{h}_{i-1} + \mathbf{b}_P$;
 $p_{\Theta}(x_i | \mathbf{h}_{i-1}) \leftarrow \text{softmax}(\ell_i)$;
 $\mathcal{L}_{\text{KMP}} += -\log p_{\Theta}(x_i | \mathbf{h}_{i-1})$;

$\mathcal{L}_{\text{KMP}} \leftarrow \frac{1}{|\mathcal{M}|} \mathcal{L}_{\text{KMP}}$;

return $\mathcal{L}_{\text{KMP}}$;

Backpropagation and Parameter Update. Because the loss is defined at the level of the final projection, gradients propagate through both projection parameters and the entire BKA stack, refining lexical embeddings, entity embeddings, and relation embeddings in concert.

B.3 Contrastive Graph Semantic Aggregation

Contrastive learning has emerged as a powerful unsupervised paradigm for representation learning, particularly in vision and speech domains. Its core premise is to pull together embeddings that originate from different views of the same underlying sample while pushing apart embeddings from distinct samples. In the context of knowledge-graph language models, however, naive contrastive approaches face two entrenched challenges. Entities and relations seldom appear in isolation; textual paraphrases and graph augmentations induce heterogeneous perturbations.

The proposed CGSA addresses these issues by marrying graph topology-aware augmentation with a Transformer encoder that is already enriched by BKA and trained under KMP. CGSA sits on top of this encoder and imposes a clustering prior on entire sub-graph representations, ensuring that embeddings remain discriminative despite repeated dropout or textual paraphrases. Conceptually, CGSA plays the role of a semantic pressure valve: it prevents the high-capacity encoder from collapsing entity neighbourhoods into an overly tangled representation space.

B.3.1 Algorithmic Steps. Input to CGSA is a mini-batch of B samples, each being either a textual snippet or a KG sub-graph. For clarity, we call each sample s_k a semantic unit. The algorithm proceeds in two macro phases: (i) stochastic view generation and (ii) InfoNCE loss computation.

Stochastic View Generation. For every semantic unit s_k , CGSA calls an augmentation routine twice, producing $\mathbf{x}_k^{(1)}$ and $\mathbf{x}_k^{(2)}$. The augmentation operator obeys two design principles:

Algorithm 3: Contrastive Graph Semantic Aggregation (CGSA)

Input: Minibatch of subgraphs or text snippets
 $\mathcal{S} = \{s_1, \dots, s_B\}$; temperature τ ; hop threshold h ;
 BKA + KMP encoder f_{Θ} ; pooling operator $\text{Pool}(\cdot)$

Output: Contrastive loss $\mathcal{L}_{\text{CGSA}}$

$\mathcal{L}_{\text{CGSA}} \leftarrow 0$;

foreach $s_k \in \mathcal{S}$ **do**
 Generate two stochastic views by augmentation;
 $\mathbf{x}_k^{(1)} \leftarrow \text{Augment}(s_k, h)$;
 $\mathbf{x}_k^{(2)} \leftarrow \text{Augment}(s_k, h)$;
 $\mathbf{H}_k^{(L,1)} \leftarrow f_{\Theta}(\mathbf{x}_k^{(1)})$;
 $\mathbf{H}_k^{(L,2)} \leftarrow f_{\Theta}(\mathbf{x}_k^{(2)})$;
 $\mathbf{z}_k^{(1)} \leftarrow \text{Pool}(\mathbf{H}_k^{(L,1)})$;
 $\mathbf{z}_k^{(2)} \leftarrow \text{Pool}(\mathbf{H}_k^{(L,2)})$;
 $\mathbf{z}_k^{(1)} \leftarrow \mathbf{z}_k^{(1)} / \|\mathbf{z}_k^{(1)}\|$;
 $\mathbf{z}_k^{(2)} \leftarrow \mathbf{z}_k^{(2)} / \|\mathbf{z}_k^{(2)}\|$;

for $k \leftarrow 1$ **to** B **do**
 $\text{pos} \leftarrow \exp(\text{sim}(\mathbf{z}_k^{(1)}, \mathbf{z}_k^{(2)}) / \tau)$;
 $\text{den} \leftarrow 0$;
 for $\ell \leftarrow 1$ **to** B **do**
 $\text{den} += \exp(\text{sim}(\mathbf{z}_k^{(1)}, \mathbf{z}_{\ell}^{(2)}) / \tau)$;
 $\mathcal{L}_{\text{CGSA}} -= \log(\text{pos} / \text{den})$;

$\mathcal{L}_{\text{CGSA}} \leftarrow \mathcal{L}_{\text{CGSA}} / B$;

return $\mathcal{L}_{\text{CGSA}}$;

Graph Reachability: A hop threshold h limits the radius within which entities may be dropped or substituted. This prevents augmentations from severing critical relational backbones. Textual Paraphrasing: For purely textual spans, augmentation may include synonym replacement, span deletion, or entity order shuffling, with the constraint that entity co-occurrence statistics are preserved up to the second moment.

Each augmented view is then passed through the encoder f_{Θ} . Because f_{Θ} already employs BKA masks, it seamlessly integrates both local lexical features and multi-hop structural cues. The encoder returns token-level hidden states $\mathbf{H}_k^{(L,\cdot)}$, which are pooled into fixed-length vectors by a user-selected operator (e.g., mean pooling, max pooling, or [CLS] token extraction).

InfoNCE Loss Computation. For each anchor index k , the algorithm forms a positive pair $(\mathbf{z}_k^{(1)}, \mathbf{z}_k^{(2)})$ and computes their scaled similarity, labelled pos . It then sums exponentiated similarities between the anchor $\mathbf{z}_k^{(1)}$ and all second-view embeddings $\mathbf{z}_{\ell}^{(2)}$ across the mini-batch, accumulating in den . The resulting expression implements the InfoNCE objective, which can be shown to maximise a lower bound on mutual information between the two views. Averaging over the batch yields the final loss $\mathcal{L}_{\text{CGSA}}$, later combined with BKA + KMP losses during multi-task training.